Overview of Approach and Modelling Strategy:

In this project the goal was to find out whether a given customer will default next month, using demographic data (education, sex, marital status, age, etc), credit limits, payment history, patterns and billing amounts of the last six months.

Missing Value Handling:

1) **Payment status features (pay_0 to 6):**
   Here a fill forward approach was followed to basically fill with the previous known status of a customer, as they show a consistent behaviour in their habits of when they repay.
2) **Bill amount, payment amount and credit limits:**

   Median was taken here to fill in the missing values, to not skew the value too much and finding roughly an average value.

3) **Demographics (Education, sex, marital status, age, etc):**
   Mode was taken as here taking the most common value is apt.

Feature Engineering:

1) **Payment delays:**
   Max, min, mean and std dev of delays to capture the variability in the extent of late payments.
2) **Bill and Payment amounts:**
   Avg change month wise and the std dev across 6 months.
3) **Credit Utilization:**
   Ratio of avg bill amt to avg credit limit which tells us how much of their available credit a customer typically uses.

Modelling Strategy:

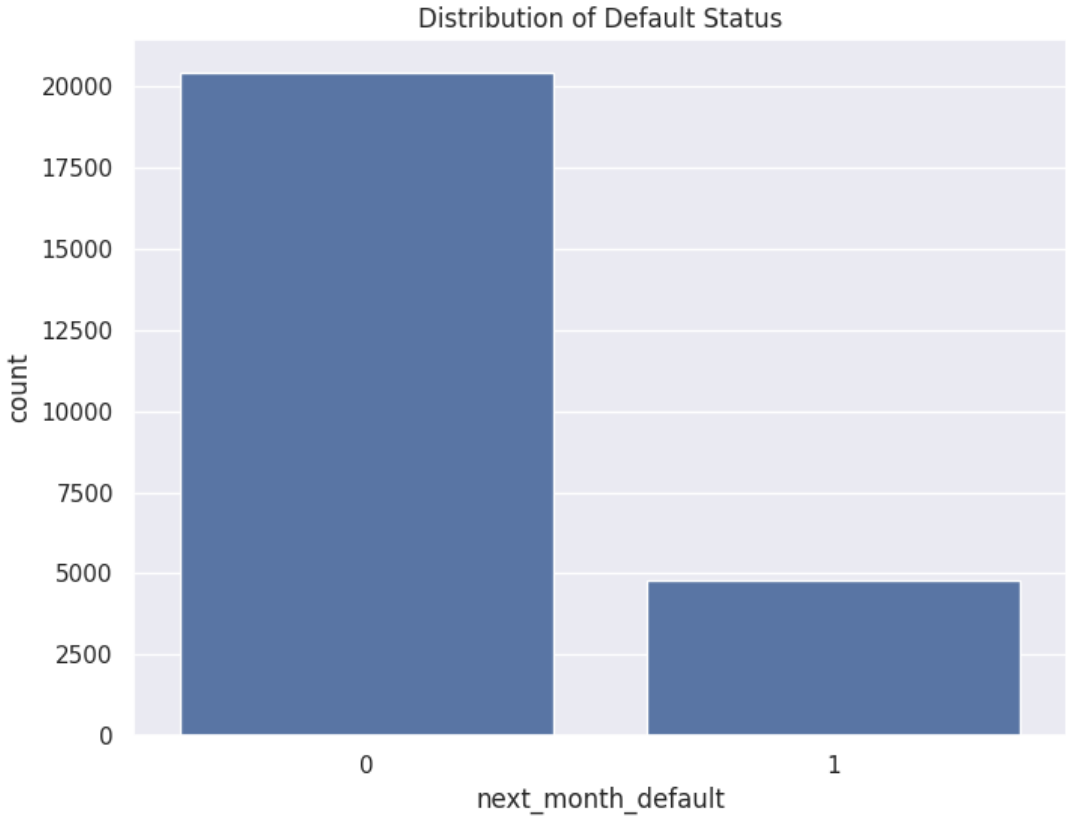
Various models were trained using the following algorithms :

- logistic regression
- decision trees
- random forest, XGBoost, LightGBM
- a majority-vote ensemble

The priority is to maximize the recall(TP/TP+FN) than the precision(TP/TP+FP), cuz we need to minimize the number of false negatives(FN) than the false positives(FP), as classifying a defaulter as a non defaulter would incur a far greater loss than classifying a non defaulter as a defaulter (maybe a proper apology). The classification threshold is optimized such that the F2 score is maximum which gives higher priority to recall than to precision.

EDA Findings and visualizations:

1) **Default Status (The target variable) imbalance:**



The Default Rate is 0.19039885927040837

So approx. only 19% of all data is of defaulters (target variable y=1).

This was resolved by SMOTE (synthetic minority oversampling technique), expanding the space of defaults' data (y=1) and class weights, so that more importance given to defaults than non-defaults.

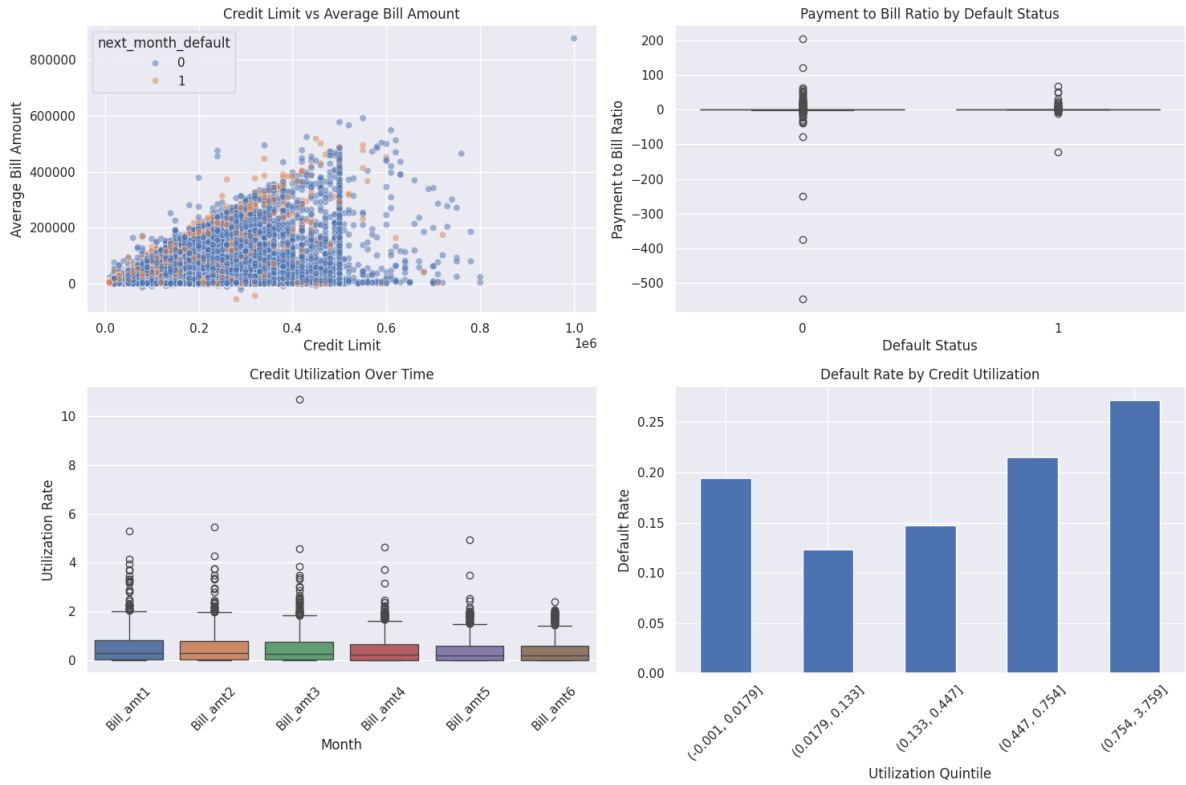
2) **Credit Utilization and billing trends:**

Defaulters are more concentrated in mid to high credit limits with high bill amounts.

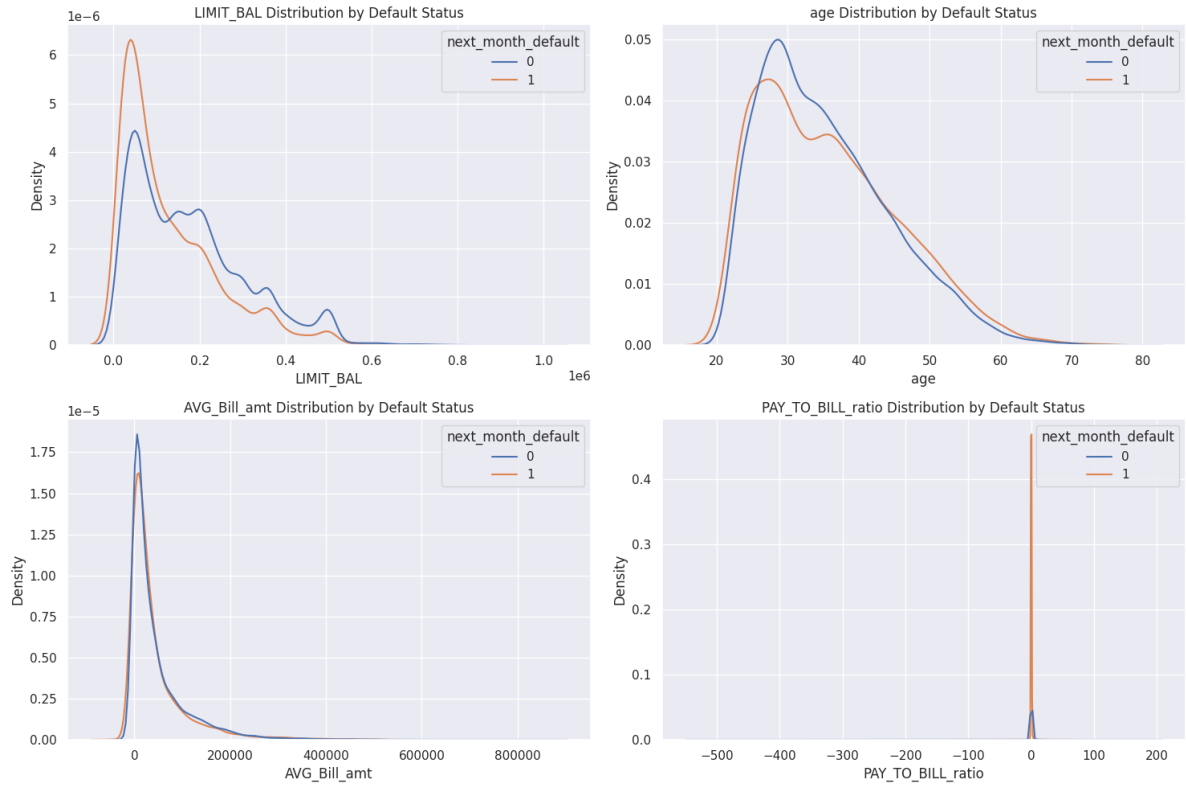
No significant difference in median payment to bill ratios between defaulters and non-defaulters.

Utilization rates remain fairly consistent across the six months.

Default rate increases with higher utilization.



3) Feature Distributions by Default Status:

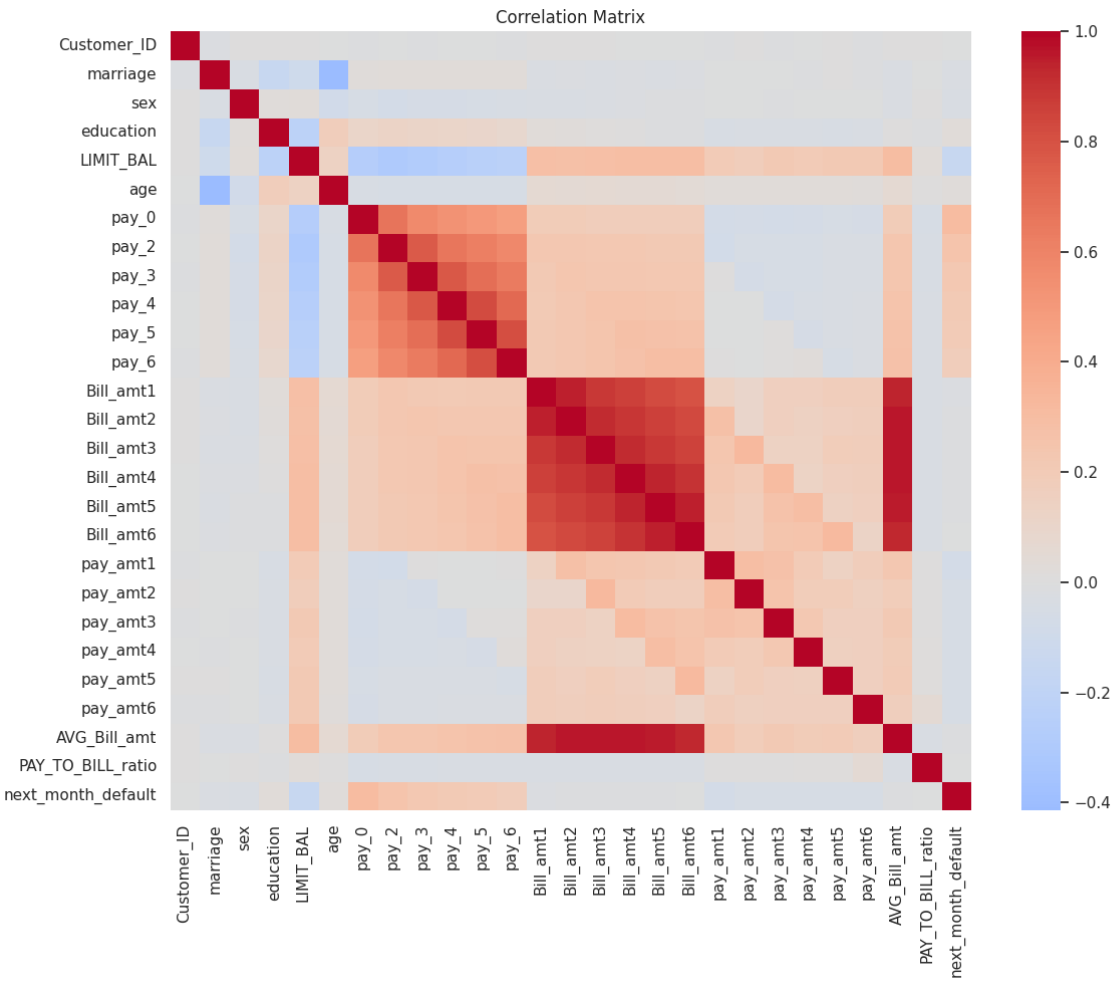


Both groups peak around lower credit limits. Defaulters have slightly lower avg credit limits than non defaulters.

Most of the users are from ages 25-35. Age is not a strong determining factor of default risk.

Defaulters have a sharp spike in pay to bill ratio while for non defaulters the curve is more spread out. So, defaulters are more likely to make minimal payments.

4) **Correlation Matrix:**



The strongest predictors are recent payment behaviours. Bill amounts, pay to bill ratios, credit limits have milder effects and marriage, sex, etc seem to be mostly non informative features.

Financial Analysis:

1) **Overdue payments:**
   Customers with pay_0 to 6 >=1 have higher probability of default, with pay_0 havi. This is because late payment,especially a recent one shows cash-flow

issues. The longer or more frequent the delay, higher the probability of no re-payment.

2) **Credit utilization:**
   Credit_utilization=avg_bill_amt/credit_limit
   If the credit utilization is high, it means a person is already using a large portion of their current limit. This leaves less buffer for new expenses which leads to higher probability of default.
3) **Credit Limit:**
   Usually ppl whom bank assesses as trustworthy who have a good history of repayment have a higher credit limit, so lower the limit of a person, more likely they are to default.
4) **Payment to Bill ratio:**
   Even if a person with large bills tends to pay on time, i.e. higher this ratio, less likely they are to default in the future.

Model Comparison:

Model	**Accuracy**	**Precision**	**Recall**	**F_2-Score**	**AUC-ROC**
Logistic Regression	0.610	0.294	0.747	0.571	0.746
Decision Tree	0.615	0.300	0.765	0.584	0.748
Random Forest	0.519	0.267	0.875	0.602	0.780
XGBoost	0.457	0.249	0.918	0.597	0.779
LightGBM	0.501	0.260	0.877	0.595	0.772
Ensemble (majority)	0.582	0.290	0.824	0.602	0.779

Out main goal is to maximize the number of defaulters we can detect, so minimum FN (high recall) even if it leads to some FP (lower precision). So, F-2 score needs to be maximized. Random Forest had the highest F-2 score of 0.602 .

Random Forest also had the highest AUC-ROC 0.780, which is a plot of true positive rate vs false positive rate.

XGBoost has the highest recall 0.918 but its precision is smaller around 0.249 . Random Forest has a slightly better precision 0.267 at nearly similar recall 0.875 .

The ensemble has same F-2 as random forest but its AUC-ROC and recall are slightly worse than it. Also, it is more complex without giving any significant improvements over random forest. So, **Random Forest** is the final selection.

Evaluation methodology:

F-2 score gives twice the importance to recall than precision i.e. more focus on minimizing the False negatives than the False positives. So, it's the most important metric.

Next is the AUC-ROC which is equivalent to the probability that a randomly chosen defaulting customer will receive a higher predicted default-probability than a randomly chosen non-defaulter. 0.5 AUC means the model is doing random guessing. Closer the AUC is to 1, the better the model is.

Among, precision and recall, more priority is given to improving the recall.

Classification cutoff:

The cutoff was determined by using five fold cross-validation. Each model was trained on four folds and tested on the remaining fold. With every customer assigned a probability, 101 thresholds were tried from 0.00 to 1.00 in steps of 0.01 to turn those probabilities into "default" or "non-default" labels. For each threshold, the F-2 score was calculated, which gives recall twice as much as importance as precision. The threshold that gave the highest overall F_2 (0.28 for the Random Forest model) was chosen as the final cutoff.

Business implications:

From a business perspective, this model helps the bank better identify high-risk customers who are likely to default on their credit card payments. By prioritizing recall and using an optimized cutoff, the system catches more true defaulters, reducing the risk of financial loss from unpaid balances. Even if this means a few non-defaulters are flagged by mistake, it's a safer trade-off for the bank. This allows for more informed decisions on credit limits, interest rates, or approval policies, ultimately improving risk management and profitability.

Summary of findings and learnings:

The six most recent payment-delay indicators (pay_0 to pay_6) explain the vast majority of variance in default probability, whereas demographic attributes contribute only marginally. To address the 19 percent default rate, we applied SMOTE oversampling on the minority class and introduced inverse-frequency class weights during training. We then performed decision-threshold tuning with the F_2 score as our objective to prioritize recall of true defaulters. Among all the models, random forest achieved the highest discrimination (AUC = 0.78) and F_2 = 0.60, and was therefore chosen.